FreshWay — Department file: Storage

Smart Logistics (EDSLOG.24) · autumn 2026

This is **your department's file**. Every department has one; you only receive your own. If you need a figure from a neighbouring department, ask them — that is what the Data Council and the Interface Agreements are for.

Alongside this file you have the **public Company dossier** (the company as a whole, the network, the cost structure, the IT landscape), which everyone has. Anything in neither document can be requested from the Executive Board with an **Information Request**; the *Board archive index* lists what exists.

Units matter. Shrink appears twice in this company: in **kilograms** against purchase volume, and in **euros** at cost. They are different numbers. Every figure below says which.

Read the KPI table with this in mind. Every KPI carries a label:

Label	Meaning
measure	not printed here on purpose — this value is in the dataset you receive in lesson 3. Measuring it is part of your work, and a department that reports its own KPI without having measured it will be asked how it knows.
N	a **norm**: what the company has set as its target. A norm is policy, not a measurement — you may argue it is wrong.
S	a **design assumption**: plausible, internally consistent, not measured. If you build a business case on it, say so.

Almost every "now" value sits below its norm. That is deliberate: a company that already meets all its targets gives you nothing to design for.

What we do. Ambient, chilled and frozen storage across three DCs: put-away, replenishment of pick faces, stock accuracy, and the cold chain inside the building. We hold the **FreshWay Energy & Capacity File** — what the grid and the buildings allow.

Volumes	Value	Unit
DC floor area, excl. EFC	**48,000** (CDC 30,000 · RDC 9,000 each) = 400 per store	m²
Freezer area	6,000 (CDC, −25 °C)	m²
Chilled area	CDC 2,000 · RDC 7,000 each	m²
Put-aways	~3,890	pallets/day
Throughput handled	311,000 cases to stores + 34,600 to the EFC	per day
Stock held in the DCs	€25 mln — fresh 1-2 days · ambient 10-14 · frozen 21	€ / days
DC energy use	**15.8 GWh/year** (electricity 11.0 · gas 4.8)	GWh
— CDC	4.65 GWh electricity · 210,000 m³ gas · 155 kWh/m²	
— RDC each	1.58 GWh electricity · 175 kWh/m²	
Refrigeration share of DC electricity	**~69** · **S** — reported, never checked against the meter data in `site_energy_hourly.csv`	%
CDC grid connection	**1.5 MVA contracted · peak ~1.06 MW · ~0.45 MVA free**	
RDC grid connection	0.6 MVA each · peak ~0.36 MW · ~0.25 MVA free	
On-site generation	16.8 GWh across DC and store roofs = 22% of electricity demand	
Pallet locations	CDC ~40,000 · RDC ~7,500 each (≈1.3 per m²)	locations · S

KPI	Now	Target
Stock accuracy	**98.6%** · S	**99.5%** · N
Temperature excursions	*measure* (D5 and D2a)	**< 5 per site per month** · N
Location utilisation	**87%** · S	**85-90%** · N — *above 92% you lose put-away flexibility, so higher is not better*

KPI	Now	Target
Energy per case handled	~0.08 kWh (DC electricity ÷ 97 mln case picks)	**0.07 kWh** · N
DC shrink	0.15% of throughput	**0.10%** · N

Systems. One WMS platform in four instances (three DCs and the EFC); a separate WCS/WES for the mechanisation in the CDC with a closed, vendor-proprietary interface; building management for refrigeration. *What they do not do:* the refrigeration system and the WMS do not talk, so we cannot relate an energy peak to a work peak; there is no real-time stock view between DCs; and we have no forward view of available electrical capacity.

People. ~595 headcount, ~480 FTE across three DCs; CDC runs 3 shifts, the RDCs 2, six days a week. Recruiting for the freezer is our hardest vacancy: absence 7.5%, turnover 22% per year, 18% agency staff and up to 30% in peak weeks. **Cost per unit:** warehousing costs €49 mln per year = **€0.50 per case** (2.2% of net revenue). A freezer hour costs 8% more than a chilled hour under the collective agreement.

Our main problems.

1. Monday morning is a wall. Everything that did not move on Sunday arrives at once, the pick faces run empty, and we solve it with agency staff rather than with a plan.
2. Refrigeration is *reported* as roughly 69% of our electricity — a figure that has been repeated for years and never checked against the meter — and we cannot say what one degree of setpoint is worth, because the cooling system and the warehouse system keep separate records.
3. Our chilled space in the regional DCs is 7,000 m² per site, dimensioned on the annual average — and in a peak the chilled share of the volume grows faster than the volume itself, because that is what people buy at Christmas.
4. **Everyone wants to electrify or automate, and the answer is no — and not only for the reason people expect.** The CDC has about 0.45 MVA of headroom; a fully electric truck fleet needs 1.2 MW more, so the shortfall is **1.3 MVA** and a new connection takes one to two years, up to ten in a congested area. But even if the power existed, there is a second limit: **time**. The charging window is 18:00-06:00 — twelve hours. What we draw in it **today** is in our own meter data and it is not the whole window, because only **8 of our 54 tractors are electric**. What it becomes at 54 is a sum nobody here has made, and it is the sum that decides the answer. So the constraint is capacity *and* time, and that pushes the answer towards smart charging, battery storage, our own generation, or simply fewer kilometres — and that last one is a logistics question, not an energy question. **Measure today first, then build the scenario on top of it; the two are different numbers and we have mixed them up before.**

What we send / receive. *To Orderpicking:* stock availability and pick-face replenishment. *To Incoming transport:* put-away confirmations and quality findings. *To E-commerce:* what is genuinely available for the fresh promise. **Data we hold but do not use:** temperature and energy readings per zone at fifteen-minute intervals, two years deep, and the ~420 mln readings a year coming off the chilled cabinets in the stores.

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*Figures come from the FreshWay figure sheet. Fields marked **measure** are in the data you receive in lesson 3. The problems in the list above are real and traceable in that data; if you cannot find them, you are looking in the wrong place or joining the wrong sets — and some of them cannot be found from your own data alone.*

> **GenAI transparency.** This department file was written with Claude (Anthropic) by the course team, based on the FreshWay case design and on verified benchmark research into Dutch grocery retail. All figures come from a separate, source-checked figure sheet; none were generated by the model. Human-AI Agency Label: **Co-Creator**. Label source: Boetje, J., & Baake, G. (2026). *Nine prototypical human-AI agency patterns* [Figure]. figshare. https://doi.org/10.6084/m9.figshare.31706884 (CC BY 4.0).
